

ernie

A daily visit that notices a bad day at home before it becomes a hospital day.

Christopher Pluntke · September 2026



Margaret is 84.

Every morning she signs Ernie's book. This morning she was slower, and nobody was there to notice.

Hypoactive delirium, the most common kind, looks like nothing at all: quieter, slower, a little withdrawn. The cause is usually cheap to fix on the day: an infection, dehydration, a new tablet. Missed, it ends in the emergency room.

60–75%

of cases unrecognised in hospital and in the ED ¹

7 M+

hospitalised US adults affected each year ²

40–100 k

lives a year, if home detection prevented delirium as well as hospital programmes do. A ceiling ⁸

Delirium is a change from your own baseline. The home never had one.

¹ Narrative review (PMC 2026); ED delirium screening (PMC 2024). ² Frontiers in Public Health 2026, MCBS 2019–21. ⁸ Witlox et al., JAMA 2010 (38.0% vs 27.5% dead at ~2 years, adjusted HR 1.95) × 1.3–2.6 M older adults a year × 30–40% preventable (HELP). Association, not proof; see sources slide.

[Start again](#)

Step 1 of 4: Signing in

Thank you, Margaret.

✓ **Done**

Lovely to see you. Here is your page in the book.



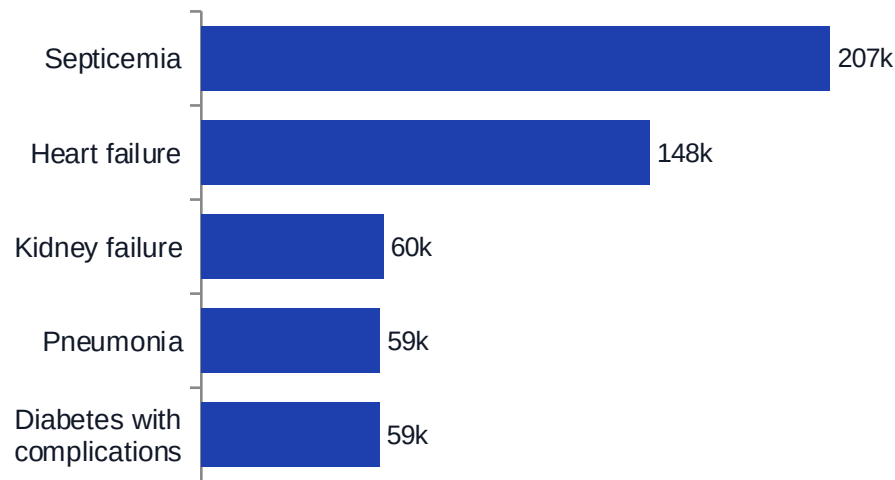
Thursday 17 September

Carry on

The gap is the 30 days at home.

Value-based care makes hospitals carry the cost of patients who come back. What sends Medicare patients back, and what a missed bad day costs.

Medicare 30-day readmissions by cause at first admission, 2020 ³



Delirium is not a diagnosis category here. It rides inside these stays and multiplies the odds of coming back by 2.6 ⁵. About 11% of Medicare hospital stays involve it ².

\$36 B

a year: 2.0 M Medicare readmissions at \$18,100 each, 17 per 100 stays ³

up to 3%

off every Medicare inpatient payment for a year when a hospital's readmissions run high. FY2022: 2,499 hospitals, \$521 M ⁴

+\$8,110

in hospital charges when a stay involves delirium, against a stay that does not ²

\$3.6–5.4 B

a year of Medicare readmissions attributable to delirium. If a quarter is avoidable, as for readmissions generally: about \$1 B ^{6 7}

³ AHRQ HCUP Statistical Brief 307, NRD 2020. ⁴ CMS HRRP; KFF, 10 Years of Hospital Readmissions Penalties (FY2022). ⁵ Community hospital cohort, n = 8,645, OR 2.60 (PMC 2019). ⁶ van Walraven et al., CMAJ 2011. ⁷ Back-of-envelope from ^{2,3,5}: upper bound, assumes the association is causal; see sources slide.

No tests. Just a visit.

She makes something every day: a signature, her own words, a finished picture. Each one is also a measurement.

Start again

Step 1 of 4: Signing in

Good morning, Margaret.

Sign my book to come in.



Any signature is fine. There is no wrong way.

Start again

Done signing

Sign the book

motor baseline · orientation

Start again

Step 2 of 4: A few questions

What day of the week is it today?

Sunday

Monday

Tuesday

Wednesday

Thursday

A few questions

her own words · latency

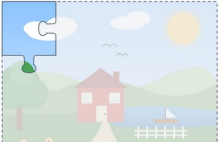
Start again

Step 3 of 4: A puzzle

Finish the picture

Drag a piece onto the picture.
Anna will help.

11 pieces left · Anna is here



Show me where this goes

Leave it for now

A puzzle, with Anna

attention · space · company

Start again

Step 4 of 4: Goodbye

Goodbye, Margaret.

✓ **Done**
That is everything for today.

Thank you for your company.
I will be here again tomorrow morning.

Start again

Goodbye

did she come · how fast

What the phone saw.

“Margaret’s signature was slower, more hesitant and a different shape than usual today.”



Her usual, and today. It looks the same. It wasn't.

What the family sees. Margaret sees a thank-you and her page in the book.

FEATURE	TODAY	USUAL	ROBUST Z
Time to first stroke	418 ms	420 ms	-0.1
Total time	4.46 s	3.06 s	+5.7 unusual
In-air time	1.07 s	267 ms	+37.8 unusual
Strokes	3	3	+0.0
Mean speed	0.650	0.778	-2.1 unusual

Nine features against her own five-day baseline: time to start, total time, pauses, strokes, speed, smoothness, size, shape.

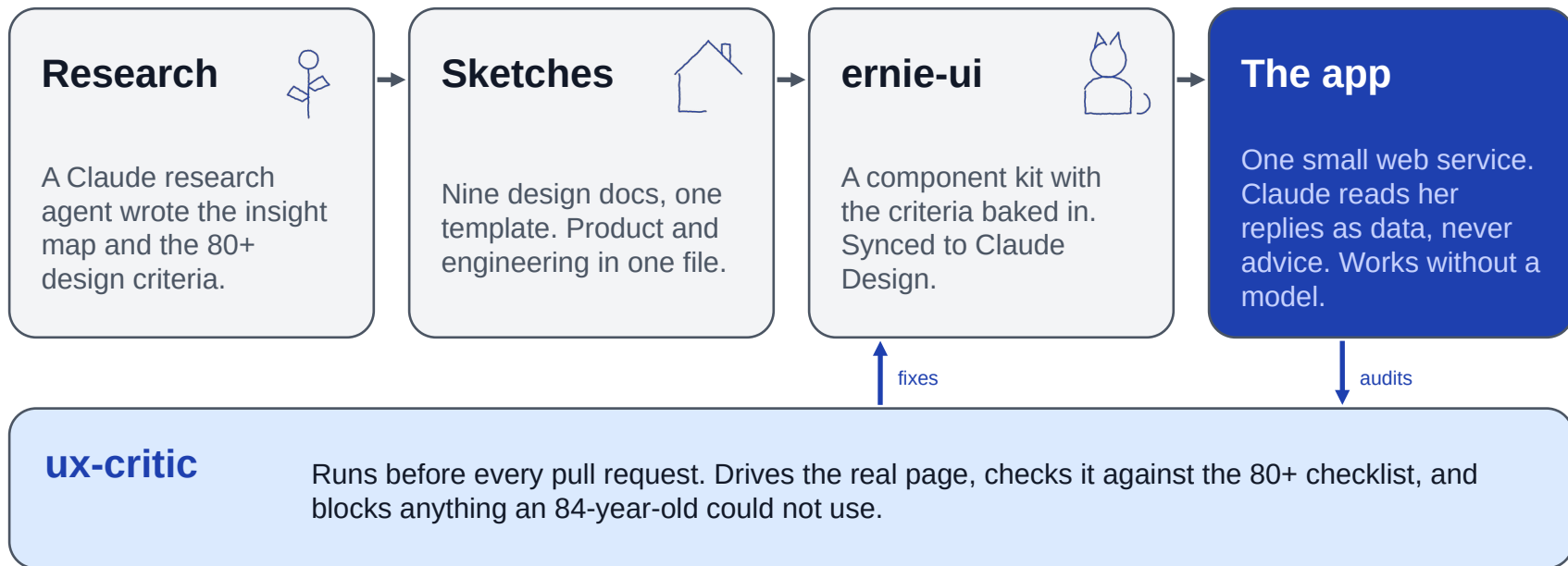
Robust z: today minus her usual, over her usual spread. Well outside on two features is a flag. Three days is a trend.

Real numbers from the demo build.



How it was built.

A pipeline with a reviewer in it.



What's next

1

Data leaves the phone

One process, one file: events, baselines, a family link.

2

Three real people, three days

Does a finger signature hold a baseline? Does a live doodle beat a check-in?

3

The explanation layer

A flag opens what changed, what it usually means, who to call before the ER.

Decision support, never a diagnosis. A human on every flag.



Synthetic people only. Not a medical device.

Sources

- 1 Delirium unrecognised in up to 60% of hospital cases and about 75% in the emergency department.
Delirium in Hospitalized Older Adults: A Narrative Review (PMC, 2026); Delirium screening in the emergency department (PMC, 2024).
- 2 Delirium affects over 7 million hospitalised US adults a year; 11% of Medicare beneficiaries had an episode; +\$8,110 hospital charges per delirium stay (95% CI \$1,860–\$14,360).
Additional hospitalization costs associated with delirium among older adults, Frontiers in Public Health, 2026 (Medicare Current Beneficiary Survey 2019–2021).
- 3 Medicare: 2,002,700 30-day readmissions in 2020, 17.0 per 100 index stays, \$36.2 B aggregate, \$18,100 average. Top causes at index admission: septicemia 207,300; heart failure 147,800; acute renal failure 60,000; pneumonia 58,900; diabetes with complications 58,700.
AHRQ HCUP Statistical Brief #307, Nationwide Readmissions Database, 2020.
- 4 HRRP reduces a hospital's Medicare inpatient payments by up to 3% for a fiscal year when 30-day readmissions for six conditions exceed the expected rate. FY2022: 2,499 of 3,139 hospitals penalised, average 0.64%, 39 at the 3% maximum, about \$521 M withheld.
CMS, Hospital Readmissions Reduction Program; KFF, 10 Years of Hospital Readmissions Penalties.
- 5 Inpatient delirium associated with 30-day readmission, adjusted OR 2.60 (95% CI 1.96–3.44); 718 delirious vs 7,927 non-delirious patients, one community hospital, 2010–2015.
Association between Inpatient Delirium and Hospital Readmission in Patients ≥65 (PMC, 2019).
- 6 Median 27% of readmissions judged avoidable across 34 studies (range 5–79%). Multicomponent prevention (HELP) cut delirium incidence from 15.0% to 9.9%.
van Walraven et al., CMAJ 2011; Inouye et al., NEJM 1999.
- 7 Back of the envelope: 11.8 M Medicare stays × 11% = 1.3 M delirium stays. With 2.0–2.6× the readmission risk and a 17% overall rate, delirium stays run 15–23 points above the rest: 200–300 k excess readmissions × \$18,100 = \$3.6–5.4 B a year. × 27% avoidable ≈ \$1.0–1.5 B. Upper bound: delirium also marks sicker patients, and no trial yet shows a home check-in reduces readmissions.
Derived from sources 2, 3 and 5.
- 8 Lives: 38.0% of older patients who had delirium were dead at ~2 years vs 27.5% of controls (adjusted HR 1.95, 95% CI 1.51–2.52; 7 studies). That 10-point excess on 1.3–2.6 M older adults a year is 130–270 k deaths associated with delirium. If home detection prevented delirium as well as in-hospital programmes (30–40%) and deaths fell with it: 40–100 k a year. Association only; delirium also marks frailty, and no trial shows a home check-in saves lives.
Witlox et al., JAMA 2010; Inouye et al., NEJM 1999; sources 2 and 3.